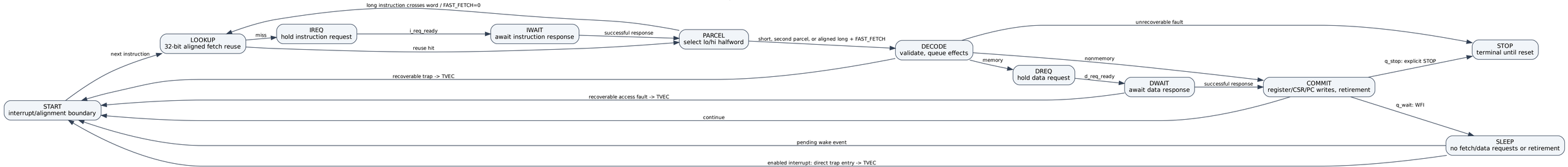
MICA 2.0 | transactional controller, MICA2=1
START
interrupt/alignment boundary
LOOKUP
32-bit aligned fetch reuse
IREQ
hold instruction request
IWAIT
await instruction response
PARCEL
select lo/hi halfword
DECODE
validate, queue effects
DREQ
hold data request
DWAIT
await data response
COMMIT
register/CSR/PC writes, retirement
STOP
terminal until reset
SLEEP
no fetch/data requests or retirement
next instruction
miss
i_req_ready
successful response
reuse hit
long instruction crosses word / FAST_FETCH=0
short, second parcel, or aligned long + FAST_FETCH
unrecoverable fault
memory
nonmemory
d_req_ready
successful response
q_stop: explicit STOP
q_wait: WFI
recoverable trap -> TVEC
recoverable access fault -> TVEC
continue
pending wake event
enabled interrupt: direct trap entry -> TVEC